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From multivariate to functional data analysis: Fundamentals, recent developments, and emerging areas

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ABSTRACT

Functional data analysis (FDA), which is a branch of statistics on modeling infinite dimensional random vectors resided in functional spaces, has become a major research area for *Journal of Multivariate Analysis*. We review some fundamental concepts of FDA, their origins and connections from multivariate analysis, and some of its recent developments, including multi-level functional data analysis, high-dimensional functional regression, and dependent functional data analysis. We also discuss the impact of these new methodology developments on genetics, plant science, wearable device data analysis, image data analysis, and business analytics. Two real data examples are provided to motivate our discussions.

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1. Introduction

In celebrating the 50 year anniversary of *Journal of Multivariate Analysis* (JMVA), we provide a review on recent developments of functional data analysis (FDA), an important research area within the scope of the journal. Data considered in FDA include curves and images, which are commonly viewed as infinite-dimensional random vectors in a functional space [47]. Nonparametric smoothing techniques are also routinely used in FDA to recover functions from discrete and noisy measurements. JMVA has a long history of publishing cutting-edge statistical theories and methods in the aforementioned areas. An early JMVA paper by [23] studied asymptotic properties of principal component analysis for a vector of random functions, including almost sure consistency and asymptotic normality, which paved the road for functional principal component analysis (FPCA), the most widely used technique in FDA. JMVA is also known for publishing high-quality research papers on nonparametric smoothing, such as early work on M-smoothing by [42].

While statistical theory on infinite-dimensional random variables and nonparametric smoothing methods are the pillars for modern FDA, other commonly used statistical methods at the center stage of JMVA have also been extended to functional data, including dimension reduction, penalization, canonical correlation analysis, clustering, classification, and regression. FDA has become such an active research area on JMVA that a keyword search for "functional data" on the journal website results in over 800 papers, including those collected in a recent special issue on functional data analysis in 2019. Some recent JMVA publications on FDA cover topics such as, statistical inference on the mean and

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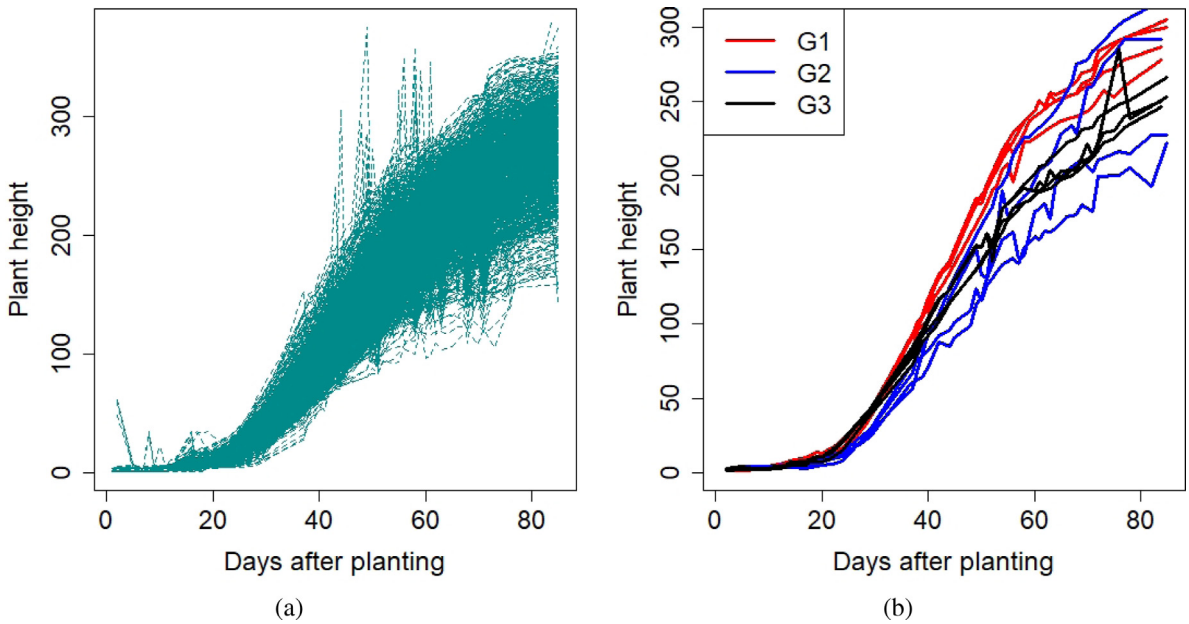

Fig. 1. Maize growth curves. (a) Growth curves for all maize; (b) Growth curves for maize of three selected genotypes.

covariance functions [13,106], nonlinear principal component analysis [97], canonical correlation analysis [44,95,128], functional linear models [15,58], functional quantile regression [119] and high-dimensional functional regression [75], just to name a few.

For comprehensive accounts of FDA theory and methods, we refer the readers to the classic textbook by Ramsay and Silverman [87], recent monographs by [46,52] and review papers by [78,107]. Hsing and Eubank [47] provided a comprehensive introduction to the theoretical foundation of FDA, especially from the perspective of operator theory. In this article, we review the fundamental theory and methods in FDA from our own relatively narrow expertise in this broad research area, including recent contributions from JMVA. We also introduce a few emerging new areas of FDA which we believe will be research active in the near future and have increasingly greater impacts on many scientific areas. We hope this paper can serve as a tutorial for researchers and domain field practitioners.

To facilitate the discussion, we first introduce two motivating data examples.

Example 1 (*TERRA-REF Data*)**.** Transportation Energy Resources from Renewable Agriculture (TERRA-REF) program aims at generating open-access reference data for the study of plant sensing, genomics, and phenomics [55]. Data were generated by a field scanner sensing platform at the University of Arizona Agricultural Center in Maricopa, Arizona. The site hosts a large field scanner with fifteen sensors, many of which are capable of capturing mm-scale images at daily to weekly intervals.

Fig. 1(a) shows the growth of 1208 maize plants based on a data set collected by the TERRA-REF program. Plant heights were extracted from raw images from 302 maize genotypes during a period of 85 days. The data are trait trajectories and have a multi-level structure among the observations, with plants nested in genotypes. The experiment also has an irregular sampling design, where on average each plant only has observations on approximately half of the days during the 85-day period. Studies of trait processes and their associations with genetic variations are of great interest to plant scientists [74,115], and those findings are important to understand the underlying biological mechanism of plant growth and produce better lines in plant breeding. Fig. 1(b) shows the growth of plants from three selected genotypes. Notice that the growth patterns are obviously different across genotypes: plants with genotype G1 grow faster than plants with genotype G3; the growth of plants with genotype G2 has more variation than the ones with genotype G3. These growth curves can be modeled as functional data with a plant-genotype nested hierarchical structure.

Example 2 (*Zillow Real Estate Data*)**.** Zillow is an online real estate company that publishes real estate data for research purposes for all major cities in the U.S. (https://www.zillow.com/research). A variable of particular interest in economics and social sciences [51] is the "home price-to-rent ratio", defined as the ratio of residential real estate price to the annual rent. It has strong relationships with market fundamentals, and has been widely used as an indicator for housing market

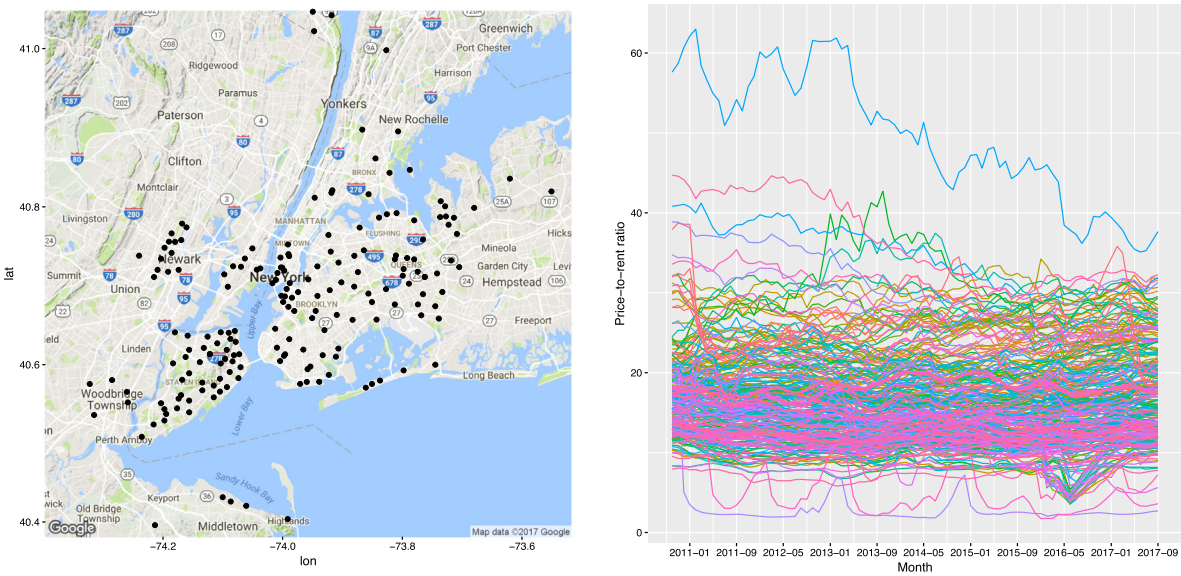

Fig. 2. New York City Zillow price-to-rent ratio data. (a) Google Map of the New York Metropolitan Area and the locations of the 278 neighborhoods; (b) A collection of home price-to-rent ratio curves during October 2010 – September 2017, discriminated by colors.

bubbles. The home price-to-rent ratio in most major U.S. cities fluctuates between 15 and 25, and a higher ratio signifies a housing bubble. This variable is updated monthly for geographical units called "neighborhoods" defined by Zillow.

We focus on New York–Newark–Jersey City Metropolitan Statistical Area defined by the U.S. Office of Management and Budget, including a population of 20.2 million people by the 2015 U.S. Census. There are 278 neighborhoods identified in the New York metropolitan area, the locations of which are illustrated in Fig. 2(a) obtained by Google Maps API. The definition of the neighborhoods can be found at https://www.zillow.com/howto/api/neighborhood-boundaries.htm. The data consist of monthly records of median home price-to-rent ratios for each neighborhood from October 2010 to September 2017. The longitudinal trajectories of price-rent ratio curves are shown in Fig. 2(b), which we model as functional data. A similar data set was analyzed by [123].

The rest of the paper is organized as follows. In Section 2, we introduce the foundations of FDA and describe its connection to multivariate analysis. In Sections 3 to 5, we summarize the state-of-the-art methods and theory for multi-level functional data analysis, functional regression with high-dimensional covariates, and dependent functional data, respectively. Some concluding remarks are provided in Section 6.

2. Foundations of FDA and their connection to MVA

In functional data analysis, we consider data consist of curves or images that are realizations of a stochastic process. Let $X_1, \ldots, X_n$ be independent and identically distributed (i.i.d.) copies of a stochastic process $X(t)$ defined on a compact domain $\mathcal{T}$. For curve data, $\mathcal{T}$ is usually a continuous interval while, for image data, $\mathcal{T}$ can be a compact closed domain in $\mathbb{R}^2$. One can define the mean and covariance function of $X(t)$ as

$$\mu(t) = \mathrm{E}\{X(t)\}, \quad \mathcal{C}(t_1, t_2) = \mathrm{cov}\{X(t_1), X(t_2)\}, \quad \forall t, t_1, t_2 \in \mathcal{T}.$$

We do not assume the covariance $\mathcal{C}$ to be stationary, which is fundamentally different from time series analysis. Recall that, to be a valid covariance function, $\mathcal{C}(\cdot,\cdot)$ needs to be positive semi-definite with $\sum_{j_1,j_2=1}^{k} a_{j_1} a_{j_2} \mathcal{C}(t_{j_1}, t_{j_2}) \geq 0$ for any $\{t_1, \ldots, t_k\}$ and any constants $\{a_1, \ldots, a_k\}$.

Let $L^2(\mathcal{T})$ be the functional space of all square integrable functions on $\mathcal{T}$. The covariance function defines an integral operator $T_{\mathcal{C}}$ on $L^2(\mathcal{T})$ such that

$$(T_{\mathcal{C}} f)(t) = \int_{\mathcal{T}} \mathcal{C}(s,t) f(s) ds, \quad \text{for any } f \in L^2(\mathcal{T}).$$

By the Mercer's theorem, $\mathcal{C}$ has an eigenvalue decomposition $\mathcal{C}(t_1, t_2) = \sum_{k=1}^{\infty} \omega_k \psi_k(t_1) \psi_k(t_2)$, where $\omega_1 \geq \omega_2 \geq \cdots \geq 0$ are the eigenvalues, and $\psi_k(\cdot)$'s are the corresponding eigenfunctions, also known as the functional principal components,

such that $\int_{\mathcal{T}} \psi_{k_1}(t)\psi_{k_2}(t)dt = \delta_{k_1k_2}$, where $\delta_{k_1k_2} = 1$ if $k_1 = k_2$ and 0 otherwise. By the Karhunen–Loève Theorem [47],

$$X_i(t) = \mu(t) + \sum_{k=1}^{\infty} \xi_{ik}\psi_k(t), \quad t \in \mathcal{T}, \tag{1}$$

where the sum converges uniformly in t, and $\xi_{ik} = \int_{\mathcal{T}}\{X_i(t) - \mu(t)\}\psi_k(t)dt$ are the functional principal component (FPC) scores such that

$$\mathrm{E}(\xi_{ik}) = 0, \ \forall k \in \mathbb{N}, \quad \mathrm{E}(\xi_{ik_1}\xi_{ik_2}) = \delta_{k_1k_2}\omega_{k_1}, \quad \forall k_1, k_2 \in \mathbb{N}.$$

The Karhunen–Loève expansion in (1) is the theoretical foundation for functional principal component analysis (FPCA). The leading principal components $\{(\omega_k, \psi_k)\}$ explain the maximal modes of variation in the data. The principal component scores $\{\xi_{ik}\}$ can be considered as subject-specific random effects on the direction of the principal components, and they explain how one curve or image deviates from the population mean. These quantities play crucial roles in the visualization, interpretation, and inference in functional data analysis.

The eigenvalues and eigenfunctions can be estimated by a spectral decomposition of the sample covariance function

$$\widehat{\mathcal{C}}(t_1, t_2) = \frac{1}{n}\sum_{i=1}^{n}\{X_i(t_1) - \bar{X}(t_1)\}\{X_i(t_2) - \bar{X}(t_2)\} = \sum_k \widehat{\omega}_k\widehat{\psi}_k(t_1)\widehat{\psi}_k(t_2),$$

where $\bar{X}(t) = (1/n)\sum_{i=1}^{n} X_i(t)$. One can also estimate the sample principal component scores by $\widehat{\xi}_{ik} = \int_{\mathcal{T}}\{X_i(t) - \bar{X}(t)\}\widehat{\psi}_k(t)dt$. Hall and Hosseini-Nasab [39] developed asymptotic expansions for the sample eigenvalues and sample eigenfunctions, which are very useful in understanding the randomness and variability in these estimators. For the purpose of dimension reduction, the number of principal components in (1) is always truncated at a finite number, which can be selected using various methods, such as the proportion of variance explained (PVE) method and the cross-validation approach [118]. Li et al. [63] proposed various information criteria to choose the number of FPCs. The sample FPC scores are widely used as predictors in linear or nonlinear regression models to predict other variables of interest [12,68,111].

In real life, we only have discrete measurements on the functions and the measurements are subject to measurement error. Suppose the observed data are

$$Y_{ij} = X_i(t_{ij}) + \epsilon_{ij}, \quad j \in \{1, \ldots, n_i\}, \tag{2}$$

where ϵ_{ij} are i.i.d. random errors with mean 0 and variance σ_ϵ^2. Note that the observation points $\{t_{ij}\}$ and the number of observations n_i in (2) are not required to be the same across different functions. Based on the number of observations on each curve, functional data are typically classified as sparse functional data [118], where n_i's are bounded by a finite number, or dense functional data [40], where $\min_i n_i \to \infty$.

Nonparametric smoothing methods have been proposed to estimate various functional parameters (such as the mean function, the covariance function, and functional principal components), including kernel methods [118] and spline methods [14]. Hall et al. [40] showed that the convergence rates for various functional estimators were different under sparse and dense functional data. Ref. [109] also showed that nonparametric hypothesis tests had different properties under sparse and dense functional data, in terms of asymptotic null distribution and power. However, sparse and dense functional data are asymptotic concepts, which are not clearly defined in any practical contexts. A lot of recent research efforts focused on developing unified estimation and inference strategies for all types of functional data [59,109,124].

3. Multi-level functional data analysis

3.1. Introduction to multi-level functional data analysis

The basic concepts of uni-level functional data analysis described in Section 2 are based on a fundamental assumption that the underlying functions are i.i.d. copies of a stochastic process. However, the i.i.d. assumption may not always hold in real life. In Fig. 1(a), the 1208 maize plants are samples from 302 maize genotypes, with 4 plants for each genotype. The data have a multi-level structure, with plants nested in genotypes. Fig. 1(b) illustrates different growth characteristics of plants with different genotypes. In the literature, multi-level functional data analysis (MFDA) is often used for analyzing data of such characteristics.

Multi-level functional data analysis has been widely used in the applications of many fields. Di et al. [24] adopted MFDA to study electroencephalographic (EEG) curves from sleep heart health studies where hospital visits of patients were nested in subjects. Zipunnikov et al. [129] studied data obtained using Magnetic Resonance Imaging (MRI) with a similar multi-level structure as in [24]. Li et al. [60] and Xiao et al. [112] studied physical activities that were measured over time using wearable devices, where the activity curves have a hierarchical structure with days nested in subjects. Motivated by a plant science root image study, Xu et al. [113] studied the bending process of seedling roots in response to gravity and the phenotypic data have a three-level nested hierarchical structure, with seeds nested in groups and groups nested in genotypes. Inspired by a scientific experiment on spectral backscatter from long-range infrared light detection and ranging data, a methodology for the analysis of multi-level binary data observed over time was proposed in [94].

Shou et al. [96] analyzed functional pitch linguistic data from a phonetic study where syllables were nested in words and words were nested in subjects.

In this section, we focus on the review of recent research in the area of multi-level functional data analysis. The TERRA-REF data in Example 1 are used to motivate the framework of our review. The motivating data have a two-level nest structure, where plants are nested in genotypes. This could be generalized to more complex structures. For examples, [60,113] investigated functional data with a three-level structure and [96] considered models with crossed or structured design. Also, our review is from a frequentist perspective, for multi-level functional data analysis in the Bayesian framework, see [56,79,80].

3.2. Multi-level functional principal component analysis

In this section, a superscript $^{[\ell]}$ is used to denote the ℓth level of the hierarchy for $\ell \in \{1, 2\}$. The genotype level and the first level are used interchangeably and the plant level is used interchangeably with the second level. The nested structure notation used by [10] is adopted for the multi-level modeling of the trait trajectories. Let $g(\cdot)$ be an index function such that $g(i)$ represents the genotype of the ith plant, $i \in \{1, \ldots, n\}$ and $g(i) \in \{1, \ldots, n_G\}$, where n_G is the total number of genotypes. The true trait trajectory of the ith plant is represented by a smooth and square-integrable random process $X_i(t)$, $t \in \mathcal{T}$, with an additive form

$$X_i(t) = \mu(t) + W^{[1]}_{g(i)}(t) + W^{[2]}_i(t), \tag{3}$$

where $\mu(t)$ is an unknown overall mean function of $X_i(t)$, and $W^{[1]}_{g(i)}(t)$ and $W^{[2]}_i(t)$ are zero-mean random processes representing functional random effects associated with the $g(i)$th genotype and the ith plant, respectively. Assume that $\{W^{[1]}_g(t)\}_{g=1}^{n_G}$ and $\{W^{[2]}_i(t)\}_{i=1}^{n}$ are independent and denote the covariance function of the random process $W^{[\ell]}(t)$ as $\mathcal{C}^{[\ell]}(t_1, t_2) = \mathrm{cov}\{W^{[\ell]}(t_1), W^{[\ell]}(t_2)\}$, $\ell \in \{1, 2\}$. If the trajectories in (3) are fully observed or the sampling time points are reasonably dense and equally spaced for each plant, method of moments estimators can be easily constructed for the estimation of $\mu(t)$ and $\mathcal{C}^{[\ell]}(t_1, t_2)$ [24].

In a large scale high-throughput phenotyping experiment, the extracted plant traits are subject to measurement errors [115], so we write the observed trait trajectory as, $Y_i(t) = X_i(t) + \epsilon_i(t)$, where $\epsilon_i(t)$ is a white noise measurement error with variance σ^2_ϵ. The images used for extracting plant traits are often taken on different dates for different plants. To reflect the possible irregular sampling design, let Y_{ij} be the jth observation on trajectory X_i at time t_{ij} for $i \in \{1, \ldots, n\}$ and $j \in \{1, \ldots, n_i\}$, where n_i is the total number of time points for the ith plant. The model for the observed trajectories is

$$Y_{ij} = \mu(t_{ij}) + W^{[1]}_{g(i)}(t_{ij}) + W^{[2]}_i(t_{ij}) + \epsilon_{ij}, \tag{4}$$

where $\epsilon_{ij} = \epsilon_i(t_{ij})$ are measurement errors.

The first step in MFDA is to estimate the mean function $\mu(t)$ using a smoothing method, e.g., local polynomial smoothing [28] or penalized spline smoothing [91]. When local polynomial smoothing is adopted, cross-validation method is often used to choose the bandwidth [118] and when penalized spline smoothing is used, generalized cross validation (GCV) [105] is usually used to select the tuning parameter. Denote the resulting estimate of the mean function as $\widehat{\mu}(t)$ and define the residuals as $Y_{ij} - \widehat{\mu}(t_{ij})$ for $i \in \{1, \ldots, n\}$ and $j \in \{1, \ldots, n_i\}$. The second step is to estimate the covariance functions $\mathcal{C}^{[\ell]}(t_1, t_2)$, $\ell \in \{1, 2\}$. The covariance functions were not obtained directly but were derived as linear transformations of smooth surfaces based on selected products of residuals using similar smoothing methods [24,114]. For example, [118] used a local quadratic regression to obtain a smooth estimate of the covariance surface, [113] used tensor-product spline smoothing, and [106] adopted polynomial spline estimation. By the Mercer's theorem, the covariance functions $\mathcal{C}^{[\ell]}(t_1, t_2)$, $\ell \in \{1, 2\}$, are positive semidefinite with spectral decomposition $\mathcal{C}^{[\ell]}(t_1, t_2) = \sum_{k=1}^{\infty} \omega^{[\ell]}_k \psi^{[\ell]}_k(t_1)\psi^{[\ell]}_k(t_2)$, where $\omega^{[\ell]}_k$ are nonincreasing eigenvalues of $\mathcal{C}^{[\ell]}$ such that $\omega^{[\ell]}_1 \geq \omega^{[\ell]}_2 \geq \cdots \geq 0$, and $\psi^{[\ell]}_k$ are the corresponding orthonormal eigenfunctions, i.e., $\int_{\mathcal{T}} \psi^{[\ell]}_{k_1}(t), \psi^{[\ell]}_{k_2}(t)\mathrm{d}t$ equals to one if $k_1 = k_2$ and equals to zero otherwise. The infinite sum is truncated and approximated by a finite sum in practice and the details of truncation methods are discussed in Section 3.3. As a third step, $\widehat{\omega}^{[\ell]}_k$ and $\widehat{\psi}^{[\ell]}_k(t)$ are solved using the traditional eigenvalue decomposition [87]. The negative eigenvalues of $\widehat{\mathcal{C}}^{[\ell]}(t_1, t_2)$ are trimmed to zero to ensure that the estimated covariance functions $\widehat{\mathcal{C}}^{[\ell]}(t_1, t_2)$, $\ell \in \{1, 2\}$, are positive semidefinite [41,118].

The final step in multi-level functional principal component analysis is the estimation of the functional principal component scores of $W^{[\ell]}(t)$ for $\ell \in \{1, 2\}$. If n_i is sufficiently large for each plant, FPC scores $\xi^{[\ell]}_k = \int_{\mathcal{T}} W^{[\ell]}(t)\psi^{[\ell]}_k(t)\,\mathrm{d}t$ can often be estimated well using numerical integration [87]. If only a few observations are available for each plant, numerical integration may perform poorly. Yao et al. [118] proposed the method of principal components analysis through conditional expectation to solve this problem for uni-level functional data and the idea can be generalized to the analysis of multi-level functional data. Assuming that $W^{[\ell]}(t)$ is truncated at the order $K^{[\ell]}$, by the Karhunen–Loève expansion, (4) can be approximated by

$$Y_{ij} = \mu(t_{ij}) + \sum_{k=1}^{K^{[1]}} \xi^{[1]}_{g(i),k}\psi^{[1]}_k(t_{ij}) + \sum_{k=1}^{K^{[2]}} \xi^{[2]}_{i,k}\psi^{[2]}_k(t_{ij}) + \epsilon_{ij}. \tag{5}$$

Notice that (5) is a linear mixed model, so the best linear unbiased prediction (BLUP) can be used to estimate the FPC scores [118]. The advantage of this method is that it borrows information from observations made on all plants. It also provides the best prediction under Gaussian assumptions [118] and is robust even if the Gaussian assumption is violated [63]. The FPC scores are often used for further analysis, such as multi-level clustering analysis [93]. The penalized spline smoothing method [91] works well under the working independence assumption, but this assumption is violated if data are multi-level, so an iterative decorrelation algorithm is recommended to improve estimation efficiency [113,117]. The prediction of the trajectory for each genotype or each plant is obtained by replacing the mean function, eigenfunctions, and FPC scores in (5) with their corresponding estimates [114].

3.3. Model selection

In practice, the Karhunen–Loève expansion (5) is often truncated at finite orders, $K^{[1]}$ and $K^{[2]}$. The orders play an important role in multi-level functional data analysis and need to be chosen carefully. A commonly used method in the principal component analysis of MVA is the proportion of variance explained (PVE) method, which chooses a few leading components that explain a subjectively chosen level of variation, such as 85% or 90%. This method is also widely used in functional data analysis due to its simplicity [22,24,60]. However, an obvious disadvantage of the PVE method is that it is not a data-driven method, so it may select the number of FPCs consistently higher or lower than the truth [113]. The potential hazard of selecting the number of FPCs incorrectly includes the loss of power in hypothesis testing [113] and the increase of prediction error for trajectories prediction [114].

Data-driven approaches based on cross-validation can also be used to choose the number of FPCs. For example, cross-validation scores were calculated based on the one-curve-leave-out prediction error for uni-level functional data [90,118] and for multi-level functional data [25]. However, the criteria based on cross-validation are very computationally intensive and simulation studies have shown that they often choose too many principal components [114]. For uni-level functional data, [63] proposed a data-driven method to decide the dimension of Karhunen–Loève expansion using conditional Akaike Information Criterion (AIC) and showed that the method could select the number of FPCs correctly under some regularity conditions. The AIC method was extended to the selection of the number of principal components for multi-level functional data analysis with irregular and possible sparse design in [114]. Since $W^{[1]}_{g(i)}(t)$ and $W^{[2]}_{i}(t)$ are independent processes, a conditional AIC is obtained by penalizing the number of estimated random effects as follows

$$\mathrm{AIC}(K^{[1]}, K^{[2]}) = -2\widehat{\ell}_n + 2(n_G K^{[1]} + nK^{[2]}), \tag{6}$$

where $\widehat{\ell}_n$ is the estimated conditional log-likelihood of the observed data given the principal component scores under Gaussian assumption. The estimated number of principal components $\widehat{K}^{[\ell]}$ are the integers that minimize (6), which can be obtained using a two-dimensional grid search algorithm. Usually, we expect the Karhunen–Loève expansion is truncated to a relatively small number of orders (e.g., within 10 or 5), so a small upper bound could be used to improve computational efficiency in the grid search algorithm [113]. Simulation studies have shown that the conditional AIC method performed well in terms of selecting the number of principal components correctly if the underlying Karhunen–Loève expansion is actually of finite orders [63,113].

3.4. Hypothesis testing

Motivated by [60,125,126] developed a model with three-level nested structures and treated the mean and random structures as smooth curves measured at various time points. The association structure of the three-level functional data was described through random curves, which were summarized using a few important principal components. Similar to [60,125] adopted penalized splines to model the fixed mean curves and the random principal component curves, and cast the proposed model into a mixed-effects model framework for model fitting, prediction, and hypothesis testing. Different from the iterative algorithm proposed by [60,117] proposed an iterative algorithm with the Expectation/Conditional Maximization Either (ECME) version of the EM algorithm [71] to estimate the model parameters. Let $\boldsymbol{\beta}$ be a vector of fixed effect parameters in the mean function. To test whether some linear contrasts of the fixed effect mean functions were zero, [60] proposed a Wald test approach with the null hypothesis, $\boldsymbol{C\beta} = \mathbf{0}$, where $\boldsymbol{C}$ was a known matrix. Simulation studies in [60] showed that a Wald statistic based on penalized likelihood method systematically led to a liberal test, but the one based on an unpenalized likelihood could achieve an acceptable test level.

Xu et al. [113] proposed a model with a three-level structure to model the processes of seedling roots bending over time in response to gravity, which were recorded using digital cameras. The functional phenotypes were collected from seedlings representing a large variety of genotypes and had a seed-group-genotype three-level nested hierarchical structure [83]. In their work, penalized splines were used for smoothing, BLUP was adopted for the prediction of FPC scores, and the conditional AIC was used to choose the number of principal components. Since the seeds were imaged on different lunar days, an important scientific question was whether there were lunar effects on root bending. To answer this question, two models were considered: the full model, where the mean bending function was a bivariate function that depended on both the recording time point, t, during the measurement process and the lunar day of measurement, s, and the reduced model, where the mean function was univariate and only depended on the recording time point, t. Xu et al. [113] proposed three versions of the generalized likelihood ratio (GLR) test statistics [30,102], which were based

on the marginal likelihood, conditional likelihood, and working independence, respectively. A permutation strategy was used to estimate the null distribution of these test statistics. Simulation studies showed that all three tests could achieve the nominal size of the tests if the number of principal components was selected correctly and the likelihood-based tests had higher power than the test based on working independence [113].

3.5. Other related topics

The framework described in Section 3.2 is limited to continuous response variables (e.g., the plant height). Motivated by a physical activity study, [62] proposed a new methodology for modeling paired functional data that consisted of simultaneous measurements of a continuous and a binary variable. Serban et al. [94] proposed methods for the analysis of multi-level binary data observed longitudinally, where the correlation was modeled among the repeated longitudinal measurements. There has also been some work on image data analysis using multivariate functional data methods. [129] considered the analysis of brain MRI data, where the response variable was a brain image composed of roughly three million measurements [3]. Motivated by [24,129] proposed a new approach to address the challenges arising from the high-dimensional response variable in multi-level functional principal component analysis.

In plant sciences, the study of trait processes and their derivatives are important for scientific findings [74]. For uni-level functional data, [22,70,81] proposed methods to recover derivatives of functional trajectories. Inspired by [70,114] proposed to recover plant trait derivative processes using the derivatives of eigenequations based on a two-level functional principal component analysis. The authors also proposed a new functional measure for the broad-sense heritability [104], which was estimated nonparametrically as a smooth function of time.

In the review above, we mainly focus on the work of multi-level functional principal component analysis. Generalized multi-level functional linear models were introduced in [18], where the response variable was continuous or discrete and the covariate had a multi-level functional structure. Both a two-stage frequentist approach and a joint Bayesian analysis approach were proposed for statistical inference. Generalized multi-level function-on-scalar regression was introduced in [36], where the response followed an exponential family distribution and was functional with a multi-level structure. The regression models in [18,36] also allowed low dimensional time-independent covariates. For the work on regression models with high dimensional covariates, see Section 4.

Spatial dependence may also exist in multi-level functional data analysis. Motivated by data obtained from a modern colon carcinogenesis scientific experiment, [98] proposed a methodological framework for the analysis of multi-level functional data, where the functions at the lowest level of the hierarchy were spatially correlated. A computationally efficient algorithm was also proposed to handle the computation for big data sets. Inspired by data from the United States Renal Data System, [61] proposed a novel multi-level spatiotemporal functional model to investigate spatiotemporal patterns of hospitalization rates among dialysis facilities, where longitudinal hospitalizations were nested in dialysis facilities and dialysis facilities were nested in geographic regions. For the review of recent work on FDA with complex dependence structures, see Section 5.

4. Function-on-scalar regression with high dimensional covariates

With the advance in high-throughput data collection, repeated measurements, longitudinal data over time and image data from many subjects are available. Such data allow the estimation of subject-specific response curves [16,17,38,84] or surfaces [65], and the interest is on studying the association between curves (or surfaces) and the potentially high-dimensional covariates of each subject. From the TERRA-REF data described in Example 1, plants from a panel of genotypes were imaged in an automatic system continuously over their growth period, and the phenotypes of interest were extracted from the sequence of images by neural network methods [1,2]. The growth curves of each genotype can be obtained from the extracted phenotypes [108]. The goal is to identify the significant single-nucleotide polymorphisms (SNPs) that can explain the difference in growth curves. Also see functional genome-wide association study (GWAS) in [17,88] for an application in the Childhood Asthma Management Project, in [66] for an application in the Alzheimer's Disease Neuroimaging Initiative and in [7] for an application in Framingham Heart Study. Other examples include studying drug effects on patients using data from wearable devices [35], and analyzing the association between brain imaging scans and individual covariates and treatment [65]. Notice that this topic of function-on-scalar regression is related to the works on varying coefficient models [43,49,50,69].

4.1. Estimation via basis expansion for coefficients

We first introduce methods for a sequence of responses for each subject that constitute a curve on $\mathbb{R}$. Let $Y_i(t)$ be the functional response of the ith subject at t for $i \in \{1, \ldots, n\}$ and $t \in \mathcal{T}$, where $\mathcal{T}$ is a compact finite interval in $\mathbb{R}$. Let $\boldsymbol{z}_i = (Z_{i1}, \ldots, Z_{ip})^\top$ be the vector of predictors of $Y_i(t)$, where Z_{ij} denotes the jth covariate on the ith subject. The linear function-on-scalar regression model is, for $i \in \{1, \ldots, n\}$ and $t \in \mathcal{T}$,

$$Y_i(t) = \beta_0(t) + \sum_{j=1}^{p} Z_{ij}\beta_j(t) + \epsilon_i(t), \tag{7}$$

where $\beta_0(t)$ is the function of intercept, $\beta_j(t)$ is the function of regression coefficient for the jth covariate, and $\epsilon_i(\cdot)$ is the error function i.i.d. drawn from a continuous stochastic process with expectation zero and covariance function $\mathcal{C}_\epsilon(t_1, t_2) = \text{cov}\{\epsilon_i(t_1), \epsilon_i(t_2)\}$ for all $t_1, t_2 \in \mathcal{T}$. Here, the dimension of covariates, p, could be much larger than the sample size n.

Let $\{\phi_\ell(\cdot)\}_{\ell=1}^\infty$ be a set of pre-specified basis functions. The coefficient functions $\beta_j(t)$ can be expanded as $\beta_j(t) = \sum_{\ell=1}^\infty b_{j\ell}\phi_\ell(t)$, for $j \in \{0, \ldots, p\}$. Truncating the basis expansion at the order m, the functional regression model (7) can be written as

$$Y_i(t) = \sum_{\ell=1}^m b_{0\ell}\phi_\ell(t) + \sum_{j=1}^p \sum_{\ell=1}^m Z_{ij} b_{j\ell}\phi_\ell(t) + T_i(t) + \epsilon_i(t), \tag{8}$$

where $T_i(t)$ is the truncation error by cutting the expansions at m. B-splines are commonly used for approximating the varying coefficient functions $\{\beta_j(\cdot)\}$ in (8). Intuitively, larger m leads to more accurate approximations of $\{\beta_j(\cdot)\}$ but at the price of higher variation of the estimators. The choice of m to balance the truncation error $T_i(t)$ and the estimation accuracy was discussed in Section 2.1 of [7]. In the following, we assume the truncation error is small and negligible, which is a typical assumption in functional regression literature.

Suppose the measurements are collected at time points $\{t_{ik}\}_{k=1}^{n_i}$ for the ith subject. The estimation problem of the functions $\beta_j(t)$, for $j \in \{0, \ldots, p\}$, is reduced to estimating the basis coefficients $\boldsymbol{B} = (b_{j\ell})_{(p+1)\times m}$, where the objective is to minimize

$$M(\boldsymbol{B}) = \frac{1}{n}\sum_{i=1}^n \sum_{k=1}^{n_i} \left\{ Y_i(t_{ik}) - \sum_{\ell=1}^m b_{0\ell}\phi_\ell(t_{ik}) - \sum_{j=1}^p \sum_{\ell=1}^m Z_{ij} b_{j\ell}\phi_\ell(t_{ik}) \right\}^2 \tag{9}$$

with some penalties on the coefficients $\boldsymbol{B}$. This is the approach adopted in [87,89] with a roughness penalty for the smoothness of the estimated functions, in [16] with a group MCP penalty [121] for variable selection under the case of fixed number of covariates, as well as in [7,17,103] for the high-dimensional case where p can be much larger than n.

To develop optimization procedures for functional regression, the loss function $M(\boldsymbol{B})$ in (9) is usually re-expressed in a matrix form so that standard optimization approaches [27,31] for regularized regression can be directly applied. Let $\boldsymbol{\Phi}_i = \{\phi_\ell(t_{ik})\}_{n_i\times m}$ be the matrix of basis functions evaluated at the time points of the ith subject, where each row of $\boldsymbol{\Phi}_i$ corresponds to a time point evaluated by all the basis functions, and each column gives a basis function evaluated at all the time points for the ith subject. Recall that $\boldsymbol{z}_i$ is the p-dimensional vector of covariates for the ith subject. Let

$$\boldsymbol{y} = \{Y_1(t_{11}), \ldots, Y_1(t_{1n_1}), \ldots, Y_n(t_{nn_n})\}^\top, \quad \mathbf{W} = \begin{pmatrix} \boldsymbol{z}_1^\top \otimes \boldsymbol{\Phi}_1 \\ \vdots \\ \boldsymbol{z}_n^\top \otimes \boldsymbol{\Phi}_n \end{pmatrix}$$

be an N-dimensional vector of responses and an $N \times pm$ dimensional matrix of , respectively, where $N = \sum_{i=1}^n n_i$ and $\otimes$ represents the Kronecker product of two matrices. Using these matrix notations, the loss $M(\boldsymbol{B})$ can be written as $M(\boldsymbol{B}) = \|\boldsymbol{y} - \mathbf{W}\text{vec}(\boldsymbol{B}^\top)\|_2^2/n$, where $\|\cdot\|_2$ denotes the Euclidean norm of vectors and $\text{vec}(\boldsymbol{B}^\top) = (b_{01}, \ldots, b_{0m}, \ldots, b_{pm})^\top$ is the column-wise vectorization of $\boldsymbol{B}^\top$. Notice that the expression of the above matrix form can be much simplified if the measurements are made at common time points for all subjects such that $t_{ik} = t_k$ for all $i \in \{1, \ldots, n\}$. See [16] for the simplified expression in this case. The estimated coefficients for basis functions solve the classical regularized optimization problem

$$\widehat{\boldsymbol{B}} = \underset{\boldsymbol{B}}{\text{argmin}}\left\{ \|\boldsymbol{y} - \mathbf{W}\text{vec}(\boldsymbol{B}^\top)\|_2^2/(2n) + \lambda P(\boldsymbol{B}) \right\}, \tag{10}$$

where λ is a smoothing parameter, and $P(\cdot)$ is some penalty function.

Let $\boldsymbol{b}_j$ denote the transpose of the jth row of $\boldsymbol{B}$. Various penalties were considered in the literature. For example, the roughness penalty $P(\boldsymbol{B}) = \sum_{j=1}^p \boldsymbol{b}_j^\top \boldsymbol{\Omega} \boldsymbol{b}_j$ was considered in [89], where $\boldsymbol{\Omega}$ is an m by m matrix with its entry in the ℓ_1th row and ℓ_2th column defined by $\int_{\mathcal{T}} \phi_{\ell_1}^{(2)}(t)\phi_{\ell_2}^{(2)}(t)dt$, and $\phi_\ell^{(2)}(t)$ denotes the second derivative of $\phi_\ell(t)$. It functions as the L_2 penalty in ridge regression that provides a smooth estimate of each coefficient $\beta_j(\cdot)$. The group lasso penalty $P(\boldsymbol{B}) = \sum_{j=1}^p \|\boldsymbol{b}_j\|_2$ was used in [7], which imposed row-wise sparsity for the estimate $\widehat{\boldsymbol{B}}$ and hence led to zero estimates for some of the coefficient functions. Related to group lasso, [16,103] considered adaptive lasso penalty and group MCP penalty for variable selection, respectively. For those aforementioned penalties, if no close form solution exists for the optimization problem (10), the coordinate descent algorithm [31] can be applied to solve it.

Notice that the loss function $M(\boldsymbol{B})$ does not consider the dependence over time within a subject. Weighted least square loss can be applied by incorporating an estimate for the error covariance function $\mathcal{C}_\epsilon(t_1, t_2)$, see [17] and Section 2.2 in [16]. It is worthwhile to mention that instead of solving an optimization problem (10), [17] proposed a sure screen algorithm for selecting the important covariates for the functional regression (7). They considered regressing $Y_i(t)$ on each single coefficient via B-spline basis expansion, and used a weighted mean square error between the true response and its prediction using each covariate as an importance measure for variable selection.

4.2. Inference for coefficients

As (10) is essentially a linear regression problem for estimating the basis coefficients $\{b_{j\ell}\}$, under fixed covariate scenarios and a roughness (L^2) penalty for $P(\boldsymbol{B})$, standard inference procedures for linear regression can be applied for constructing confidence intervals and conducting hypothesis testing for $b_{j\ell}$, which then leads to point-wise confidence intervals for $\beta_j(t)$ and an F-test for significance of covariates [89]. Meanwhile, simultaneous confidence corridor was considered in [38] under the fixed dimensional case. For $P(\boldsymbol{B})$ being the MCP penalty, [16] showed the asymptotic normality of the estimated coefficients $\widehat{b}_{j\ell}$ under the fixed covariate setting. For high-dimensional settings, [7] showed the convergence rate for the estimated coefficient functions. However, statistical inference results were not developed.

As a different approach, under the scenario of fixed covariate and observations being measured at the same time points, [88] proposed a statistic for testing the significance of a group of covariates on the response function based on the difference of the residual sum of squares between the full model and the reduced model. Here, the residual sum of squares stands for the integration of squared residual function over time. They also discussed the extension of their test statistic to the projected response function by the estimated eigenvectors of the covariance function of responses. In this case, their test statistic can be formulated as a weighted sum of proportion of explained variability of each principal components by the covariates. However, this approach cannot be applied to the data with high-dimensional covariates. How to conduct statistical inference for varying coefficient functions under high-dimensional covariates is still an open question.

4.3. Surface-on-scalar regression

In spatial statistics and medical image analysis, the data are observed in a two-dimensional domain. This requires the extension of the curve regression (7) to surface regression, for $i \in \{1, \ldots, n\}$,

$$Y_i(s) = \beta_0(s) + \sum_{j=1}^{p} Z_{ij}\beta_j(s) + \epsilon_i(s), \tag{11}$$

where $s \in \mathbb{R}^2$ is a point in a two-dimensional space. Under this model, [127] proposed a locally weighted three-stage estimation method, and [65] developed an estimation approach by approximating the spatially varying coefficient functions $\{\beta_j(s)\}$ via bivariate spline functions over triangulation. Both works allow high-dimensional covariates, but require independent repetitions of the two-dimensional images. The sample size n in (11) needs to diverge to infinity for valid asymptotic results. It is worthwhile to mention that [120] considered a related generalized geo-additive model without repetitions ($n = 1$) but under a fixed dimensional setting. Specifically, the method proposed in [120] is similar to the estimation procedure introduced in Section 4.1 with the univariate spline expansion for $\beta_j(t)$ replaced by a bivariate spline expansion for $\beta_j(s)$, which satisfies certain spatial smoothness properties.

4.4. Functional regression in Hilbert space

For densely observed functional data, the response $Y_i(t)$ can be considered as a sample point in a function space for each subject $i \in \{1, \ldots, n\}$. For the functional linear model (7), suppose the errors $\{\epsilon_i(t)\}$ are i.i.d. random elements of a real separable Hilbert space $\mathcal{H}$ with mean zero and covariance operator $\mathcal{C}_\epsilon$, and $\{\beta_j(t)\}$ are deterministic elements of $\mathcal{H}$. To estimate the regression coefficients $\{\beta_j(t)\}$, [7] considered to minimize the objective function

$$M_{\mathcal{H}}(\beta_0, \ldots, \beta_p) = \frac{1}{2n}\sum_{i=1}^{n}\left\|Y_i - \beta_0 - \sum_{j=1}^{p} Z_{ij}\beta_j\right\|_{\mathcal{H}}^2 + \lambda\sum_{j=1}^{p}\|\beta_j\|_{\mathcal{H}} \tag{12}$$

over $\beta_j \in \mathcal{H}$, $j \in \{0, \ldots, p\}$, where $\|\cdot\|_{\mathcal{H}}$ denotes the inner product norm on $\mathcal{H}$. Here, $\sum_{j=1}^{p}\|\beta_j\|_{\mathcal{H}}$ serves as an L_1 norm for the vector of functions $(\beta_1, \ldots, \beta_p)^\top$ on the product space of $\mathcal{H}$. Theoretical results have developed to show the convergence of the estimates on $\mathcal{H}$. To solve the optimization problem on $\mathcal{H}$, [7] suggested to approximate $M_{\mathcal{H}}(\beta_0, \ldots, \beta_p)$ in (12) by

$$M_{\mathcal{H}}(\beta_0, \ldots, \beta_p) \approx \frac{1}{2n}\sum_{i=1}^{n}\sum_{\ell=1}^{m}\left\langle Y_i - \beta_0 - \sum_{j=1}^{p} Z_{ij}\beta_j, e_\ell\right\rangle_{\mathcal{H}}^2 + \lambda\sum_{j=1}^{p}\left(\sum_{\ell=1}^{m}\langle\beta_j, e_\ell\rangle_{\mathcal{H}}^2\right)^{1/2}$$

for a large m, where $\langle\cdot,\cdot\rangle_{\mathcal{H}}$ denotes the inner product on $\mathcal{H}$, and $\{e_1, e_2, \ldots\}$ are a basis of $\mathcal{H}$. This approximation allows the use of the group lasso algorithm for minimizing $M_{\mathcal{H}}(\beta_0, \ldots, \beta_p)$.

This approach provides a general methodology for estimating coefficient functions under dense functional data, which includes Reproducing Kernel Hilbert Spaces (RKHS) as a special case. As an extension, [29] considered adaptive penalty, which added a data-driven weight on $\|\beta_j\|_{\mathcal{H}}$ for each $j \in \{1, \ldots, p\}$. This is similar to the adaptive lasso approach [130] for high-dimensional scalar-on-scalar regression. The weight is chosen as $\|\tilde{\beta}_j\|_{\mathcal{H}}^{-1}$ for an initial estimator $\tilde{\beta}_j$ of β_j, so that it is small for large $\|\beta_j\|_{\mathcal{H}}$, and large for small $\|\beta_j\|_{\mathcal{H}}$. Selection consistency of their estimates has been shown under

the irrepresentable condition [11] on the design matrix and the minimum signal condition on β_j, which puts a lower bound on $\|\beta_j\|_{\mathcal{H}}$ for all nonzero β_j. Also see [75,84] for further studies on constructing flexible penalties using positive definite, self-adjoint linear operators (including kernel functions in RKHS) that can accommodate a variety of underlying assumptions about the parameters, and simultaneously select significant predictors and produce smooth estimates.

5. FDA with complex dependence structures

In many applications, functional data collected at different time points or locations are naturally correlated. There have been a lot of recent theoretical and methodological developments for dependent functional data, including multi-level functional data [18,48,113], functional time series [4,45,53], and spatially dependent functional data [37,54,67,99,122,123,126]. There has also been some work on modeling spatio-temporal point process data using a functional data analysis approach [33,57].

5.1. Spatially dependent functional data under geostatistics design

The Zillow real estate data described in Section 1 provide a good example of spatially dependent functional data, where curve data are collected from a random collection of spatial collections. These data resemble classic spatio-temporal data [20], yet highlight some major differences. Classic spatio-temporal data analysis is built on random fields theory, usually used to model measurements that can be defined in continuous space and time, such as air quality, ocean temperature, etc. Due to the lack of independent replicates, spatio-temporal models usually rely on some relaxed version of stationarity assumptions. Spatially dependent functional data, exemplified by the Zillow real estate data in Section 1, are temporal processes collected from some well-defined entities (e.g., people, real estate properties, remote sensors), which happen to be correlated spatially. These entities and the corresponding temporal processes are not necessarily well defined for all locations within a spatial domain, unlike air or water. The curves in Fig. 2(b) in the introduction section are clearly not stationary in time. Our use of spatial random field theory to model correlation among spatially dependent functional data is a modeling effort analogous to the Generalized Linear Mixed Models.

Zhang and Li [123] analyzed a similar dataset from Zillow research and proposed the following framework for spatially dependent functional data under geostatistics settings. Suppose random functions of time defined on a time domain $\mathcal{T}$ are sampled from locations in a spatial domain $\mathcal{D}_n \subseteq \mathbb{R}^2$. Let $Y_{ij} = Y(\boldsymbol{s}_i, t_{ij})$ be the discrete observation at time t_{ij} on the random curve sampled at spatial location $\boldsymbol{s}_i$, $i \in \{1, \ldots, n\}$, $j \in \{1, \ldots, n_i\}$, and assume

$$Y(\boldsymbol{s}_i, t_{ij}) = X(\boldsymbol{s}_i, t_{ij}) + U_i(t_{ij}) + \epsilon_{ij}, \tag{13}$$

where $X(\cdot, \cdot)$ is a spatio-temporal process on $\mathcal{D}_n \times \mathcal{T}$ representing a spatially correlated functional effect, $\{U_i(\cdot)\}$ are zero-mean, independent temporal processes called the functional nugget effects, and $\{\epsilon_{ij}\}$ are the independent measurement errors with $E(\epsilon_{ij}) = 0$ and $\mathrm{var}(\epsilon_{ij}) = \sigma_\epsilon^2$. The functional nugget effects, $U_i(\cdot)$, characterize local variations that are not correlated with neighboring functions, with the covariance function denoted by $\Lambda(t_1, t_2) = \mathrm{cov}\{U(t_1), U(t_2)\}$. The three model components $X(\cdot, \cdot)$, $U(\cdot)$, and ϵ are mutually independent.

Assuming that the spatial dependency is second-order stationary and isotropic, the general covariance function of $X(\boldsymbol{s}, t)$ can be written as

$$\mathcal{C}(\|\boldsymbol{s}_1 - \boldsymbol{s}_2\|, t_1, t_2) = \mathrm{cov}\{X(\boldsymbol{s}_1, t_1), X(\boldsymbol{s}_2, t_2)\},$$

for any $(\boldsymbol{s}_1, t_1), (\boldsymbol{s}_2, t_2) \in \mathcal{D}_n \times \mathcal{T}$. We consider $X_{\boldsymbol{s}}(t) = X(\boldsymbol{s}, t)$ as spatial replicates of a temporal process with a standard Karhunen–Loève expansion

$$X_{\boldsymbol{s}}(t) = \mu(t) + \textstyle\sum_{j=1}^{\infty} \xi_j(\boldsymbol{s})\psi_j(t), \tag{14}$$

where $\mu(t) = E\{X_{\boldsymbol{s}}(t)\}$ is the mean function, $\{\psi_j(\cdot)\}$ are orthonormal functions known as the principal components, and the principal component score $\xi_j(\boldsymbol{s}) = \int_{\mathcal{T}} \{X(\boldsymbol{s}, t) - \mu(t)\}\psi_j(t)dt$ is the loading of $X(\boldsymbol{s}, t)$ on the jth principal component. We assume $\{\xi_j(\boldsymbol{s})\}$ are zero-mean, second-order stationary and isotropic random fields, that are uncorrelated across different j. Spatial dependence among the functional data is induced by the dependence within each $\xi_j(\boldsymbol{s})$. Denote the spatial covariance function of $\xi_j(\boldsymbol{s})$ as $\mathcal{C}_j(\|\boldsymbol{s}_1 - \boldsymbol{s}_2\|) = \mathrm{cov}\{\xi_j(\boldsymbol{s}_1), \xi_j(\boldsymbol{s}_2)\}$, for any $\boldsymbol{s}_1, \boldsymbol{s}_2 \in \mathcal{D}_n$, then the covariance function for $X(\boldsymbol{s}, t)$ can be written as

$$\mathcal{C}(\|\boldsymbol{s}_1 - \boldsymbol{s}_2\|, t_1, t_2) = \mathrm{cov}\left\{\sum_{j=1}^{\infty} \xi_j(\boldsymbol{s}_1)\psi_j(t_1), \sum_{j=1}^{\infty} \xi_j(\boldsymbol{s}_2)\psi_j(t_2)\right\} = \sum_{j=1}^{\infty} \mathcal{C}_j(\|\boldsymbol{s}_1 - \boldsymbol{s}_2\|)\psi_j(t_1)\psi_j(t_2) \tag{15}$$

Denote $\omega_j = \mathcal{C}_j(0)$ as the marginal variance for $\xi_j(\boldsymbol{s})$, and assume the principal components are ordered according to their magnitudes such that $\omega_1 \geq \omega_2 \geq \cdots \geq 0$. It is easy to see that ω_j's and $\psi_j(\cdot)$'s are the eigenvalues and eigenfunctions of the covariance function $\mathcal{C}(0, \cdot, \cdot)$, which reveals an important connection between model (14) and the classic model (1) for independent functional data. Similar FPC expansion as (14) was also considered by [46,126] for spatially dependent functional data. By allowing different orders of FPC score to have different spatial covariances, covariance function (15)

is a "coregionalization" model [6,32], which is the sum of many separable spatio-temporal covariance functions, and it reduces to a separable structure if $\mathcal{C}_j(\cdot) = \omega_j \rho(\cdot)$ for all j where $\rho(\cdot)$ is a common correlation function. The functional nugget effect $U_i(\cdot)$, on the other hand, may have an entirely different covariance structure with different eigenvalues and eigenfunctions.

Zhang and Li [123] proposed spline-based methods to estimate the functional parameters in the spatial functional model (13)–(14), including $\mu(t)$, $\mathcal{C}(u, t_1, t_2)$, $\Lambda(t_1, t_2)$, as well as the error variance parameter σ_ϵ^2. Based on the coregionalization structural assumption (15), their three-dimensional spline covariance estimator for $\mathcal{C}(\cdot, \cdot, \cdot)$ yielded important byproducts, including nonparametric estimators of the principal components $\{\psi_j(t)\}$ and the spatial covariance functions for the FPC scores $\{\mathcal{C}_j(u)\}$. They developed asymptotic convergence rates for their proposed estimator under a general framework that combines both infill and increasing domain asymptotic paradigms, and showed that their methods and theory were applicable to both sparse and dense functional data. As their theory shows, compared with the domain size, the effect of infilling locations in a unit spatial domain only has a secondary effect on the asymptotic convergence rate of their proposed estimators. Their convergence rate for $\mathcal{C}_j(u)$ is comparable to previous work on nonparametric spatial correlation estimation [64]. They also established phase transition in the convergence rates from sparse to dense functional data, which was not previously available for spatially dependent functional data.

The simulation studies in [123] also stress the importance of modeling the functional nugget effects, which models the local characteristics that are not dependent on the neighbors. They showed that ignoring the functional nugget effects can potentially cause large biases in the FPCA estimators. They also showed that not only their FPC estimators can be used for dimension reduction for spatially dependent functional data, their estimators for the spatial covariance functions can be used for functional kriging, i.e., predicting the function at a new location where no data have been collected. Spatial prediction or kriging is a major interest in spatial statistics [100] and there had been some previous work on kriging for spatially dependent functional data. The FPCA-then-kriging two-step procedure [72,82] is to first perform the classic FPCA described in Section 2 ignoring any spatial dependence and then perform co-kriging on the estimated FPC scores by fitting parametric spatial covariance models, e.g., the Matérn models. There are several issues with this procedure: first, it does not consider functional nugget effect and may suffer from large estimation biases; second, the estimated FPC scores are contaminated with estimation errors bringing a lot of nuisance into spatial covariance estimation; third, the spatial covariance models are limited to a few parametric families which may be mis-specified. The trace kriging method [34,73] does not depend on dimension reduction (e.g., FPCA) and requires fully observed functional data without measurement error nor nugget effect. In both their simulation studies and their analysis of the Zillow real estate data, Zhang and Li [123] showed that their functional kriging method outperforms competing methods by yielding smaller prediction errors.

5.2. Spatially dependent functional data under lattice design

Data collected on a regular spatial grid are commonly referred to as lattice data or areal data [19]. A frequently used model is the conditional autoregressive model (or CAR model), where data are assumed to be conditionally independent given the neighboring data. Ref. [122] extended the CAR model to spatially dependent functional data observed on a regular lattice. They considered functional data collected on a regular lattice $\mathcal{D} = \{\boldsymbol{s}_1, \boldsymbol{s}_2, \ldots, \boldsymbol{s}_n\}$, and assume there were independent replicates for the functional lattice data. Let $Y_{ij}(t)$ be the observed function on $\boldsymbol{s}_i$ from subject j, and assume

$$Y_{ij}(t) = \boldsymbol{x}_{ij}^\top \boldsymbol{b}(t) + \boldsymbol{z}_{ij}^\top \boldsymbol{u}(t) + E_{ij}(t), \quad i \in \{1, \ldots, n\}, \quad j \in \{1, \ldots, J\}, \quad t \in \mathcal{T}, \tag{16}$$

where $\mathcal{T}$ is a common time domain, $\boldsymbol{x}_{ij}$ is a p-dimensional fixed effect covariates, $\boldsymbol{z}_{ij}$ is a m-dimensional random effect covariates, $\boldsymbol{b}(t)$ is a vector of functional fixed effects, $\boldsymbol{u}(t)$ is a m-dimensional functional random effects, and $E_{ij}(t)$ are residual curves that are spatially correlated across locations i but independent among subjects j. The introduction of functional random effect $\boldsymbol{u}(t)$ in (16) is to accommodate complicated nested, split-plot, or longitudinal designs.

Ref. [122] focused on the case when both the location $\boldsymbol{s}$ and time point t are sampled on regular grid points and assumed all functions can be expressed without loss by K-dimensional basis functions $\boldsymbol{\Phi}(t)$, such as Fourier basis, wavelets or splines. Suppose the transformed data can be expressed as

$$\boldsymbol{Y}_{ij}^* = \boldsymbol{X}_{ij}\boldsymbol{B}^* + \boldsymbol{Z}_{ij}\mathbf{U}^* + \boldsymbol{E}_{ij}^*,$$

where $\boldsymbol{Y}_{ij}^*$, $\boldsymbol{B}^*$, $\mathbf{U}^*$ and $\boldsymbol{E}_{ij}^*$ are data matrices of coefficient projecting $Y_{ij}(t)$, $\boldsymbol{b}(t)$, $\boldsymbol{u}(t)$ and $E_{ij}(t)$ onto $\boldsymbol{\Phi}(t)$. They assumed $\mathbf{U}^*$ followed a multivariate normal distribution, but spatial correlation was modeled through the dependency within $\boldsymbol{e}_{\cdot jk}^* = (e_{1jk}^*, \ldots, e_{njk}^*)^\top$ using the following CAR model

$$\boldsymbol{e}_{\cdot jk}^* \sim \text{Normal}\{\mathbf{0}, s_{jk}^*(\boldsymbol{D}_v - \rho_k^*\mathbf{V})^{-1}\}, \quad j \in \{1, \ldots, J\}, \quad k \in \{1, \ldots, K\}, \tag{17}$$

where s_{jk}^* is a variance parameter, $\mathbf{V} = (v_{i_1 i_2})_{i_1, i_2=1}^n$ is an adjacent matrix with $v_{i_1 i_2} = 1$ if $\boldsymbol{s}_{i_1}$ and $\boldsymbol{s}_{i_2}$ are neighbors, 0 otherwise, and $\boldsymbol{D}_v = \text{diag}(v_{i+})$ with $v_{i+} = \sum_{\ell=1}^n v_{i\ell}$. The correlation parameter ρ_k^* in (17) are allowed to be different across k to accommodate possible non-separable spatio-temporal covariance structure for $E_{ij}(t)$. The model is then fitted as a Bayesian hierarchical model using Markov chain Monte Carlo (MCMC). The detailed algorithm can be found in [122].

5.3. Functional data analysis approach for spatio-temporal point process data

Spatio-temporal point process data arise from many scientific fields including ecology, epidemiology, and seismology. For example, spatio-temporal point process models were used to model outbreaks of weeds [9], the incidence of foot and mouth disease [26], and occurrences of earthquakes [92]. The log-Gaussian Cox process (LGCP) [8,9,76] is one of the most widely used models for spatio-temporal point process data, where the intensity function is assumed to be a log-linear model of some latent spatio-temporal Gaussian process. Li and Guan [57] proposed a functional data analysis approach to analyze spatio-temporal point process data arising from disease surveillance. Their model was based on LGCP, where the latent temporal process at any fixed spatial location was modeled as a functional process with a standard functional principal component expansion and the latent functional principal component scores were spatially correlated. Their model can accommodate both nonparametric temporal trends and complicated spatio-temporal correlations.

Let $N(\cdot,\cdot)$ denote a spatio-temporal point process that is observed on $\mathcal{W} = \mathcal{D} \otimes \mathcal{T}$, where $\mathcal{D} \subset \mathbb{R}^2$ is a spatial domain and $\mathcal{T}$ is a time domain. Let X be an L^2 Gaussian random field on W. We assume that conditional on X, N is a Poisson process with an intensity function $\lambda(\boldsymbol{s}, t)$ given by

$$\log\{\lambda(\boldsymbol{s}, t)\} = \boldsymbol{z}^\top(\boldsymbol{s}, t)\boldsymbol{\beta} + X(\boldsymbol{s}, t), \tag{18}$$

where $\boldsymbol{z}(\boldsymbol{s}, t)$ is a d-dimensional covariate vector, and X represents spatio-temporal random effects that cannot be explained by $\boldsymbol{z}$. Consider $X(\boldsymbol{s}, t)$ as spatial replicates of an L^2 Gaussian process on $\mathcal{T}$ with a Karhunen–Loève expansion (14). Then the general covariance function of $X(\boldsymbol{s}, t)$ is

$$\mathcal{C}(\boldsymbol{s}_1, \boldsymbol{s}_2, t_1, t_2) = \mathrm{cov}\{X(\boldsymbol{s}_1, t_1), X(\boldsymbol{s}_2, t_2)\} = \sum_{j=1}^{\infty} \mathcal{C}_j(\boldsymbol{s}_1, \boldsymbol{s}_2)\psi_j(t_1)\psi_j(t_2), \tag{19}$$

where $\mathcal{C}_j(\boldsymbol{s}_1, \boldsymbol{s}_2) = \mathrm{cov}\{\xi_j(\boldsymbol{s}_1), \xi_j(\boldsymbol{s}_2)\}$. The covariance in (19) is non-separable in general, and implies that $X(\boldsymbol{s}, t)$ and hence $N(\cdot,\cdot)$ are not necessarily stationary in t.

Let $N(d\boldsymbol{s}, dt)$ denote the number of points or events in an infinitesimal window $d\boldsymbol{s} \times dt$, and let $|d\boldsymbol{s}|$ and $|dt|$ denote the volumes of $d\boldsymbol{s}$ and dt, respectively. The marginal first-order intensity function, which characterizes the probability to observe an event at a given location and time, is defined as

$$\lambda_1(\boldsymbol{s}, t) = \lim_{|d\boldsymbol{s}|, |dt| \to 0} \frac{\mathrm{E}\{N(d\boldsymbol{s}, dt)\}}{|d\boldsymbol{s}||dt|} = \mathrm{E}\left[\exp\{\boldsymbol{z}^\top(\boldsymbol{s}, t)\boldsymbol{\beta} + X(\boldsymbol{s}, t)\}\right] = \exp\{\boldsymbol{z}^\top(\boldsymbol{s}, t)\boldsymbol{\beta} + \gamma(t)\}, \tag{20}$$

where $\gamma(t) = \mu(t) + (1/2)\sum_{j=1}^{\infty} \omega_j \psi_j^2(t)$. The second-order intensity function, which characterizes the correlation within the process, is defined as

$$\begin{aligned}
\lambda_2(\boldsymbol{s}_1, \boldsymbol{s}_2, t_1, t_2) &= \lim_{\substack{|d\boldsymbol{s}_1|, |d\boldsymbol{s}_2|, \\ |dt_1|, |dt_2| \to 0}} \frac{\mathrm{E}\{N(d\boldsymbol{s}_1, dt_1)N(d\boldsymbol{s}_2, dt_2)\}}{|d\boldsymbol{s}_1||dt_1||d\boldsymbol{s}_2||dt_2|} \\
&= \mathrm{E}\left[\exp\{\boldsymbol{z}^\top(\boldsymbol{s}_1, t_1)\boldsymbol{\beta} + \boldsymbol{z}^\top(\boldsymbol{s}_2, t_2)\boldsymbol{\beta} + X(\boldsymbol{s}_1, t_1) + X(\boldsymbol{s}_2, t_2)\}\right] \\
&= \lambda_1(\boldsymbol{s}_1, t_1)\lambda_1(\boldsymbol{s}_2, t_2)\exp\{\textstyle\sum_{j=1}^{\infty}\mathcal{C}_j(\boldsymbol{s}_1, \boldsymbol{s}_2)\psi_j(t_1)\psi_j(t_2)\},
\end{aligned} \tag{21}$$

for $(\boldsymbol{s}_1, t_1) \neq (\boldsymbol{s}_2, t_2)$. Given the first-order intensity function (20) and second-order intensity functions (21), the pair correlation function [77] for the point process is

$$\mathcal{G}_2(\boldsymbol{s}_1, \boldsymbol{s}_2, t_1, t_2) = \frac{\lambda_2(\boldsymbol{s}_1, \boldsymbol{s}_2, t_1, t_2)}{\lambda_1(\boldsymbol{s}_1, t_1)\lambda_1(\boldsymbol{s}_2, t_2)} = \exp\left\{\sum_{j=1}^{\infty} \mathcal{C}_j(\boldsymbol{s}_1, \boldsymbol{s}_2)\psi_j(t_1)\psi_j(t_2)\right\}.$$

Assuming $\mathcal{C}_j(\boldsymbol{s}_1, \boldsymbol{s}_2) = \mathcal{C}_j(\boldsymbol{s}_1 - \boldsymbol{s}_2)$ is stationary, then $\mathcal{G}_2(\boldsymbol{s}_1, \boldsymbol{s}_2, t_1, t_2)$ is a function of $\boldsymbol{s}_1 - \boldsymbol{s}_2$ and (t_1, t_2). Hence, the point process is second-order intensity reweighted stationary in space [5].

To fit the model (18), Li and Guan [57] proposed Poisson maximum likelihood and composite likelihood methods based on spline approximations to estimate the mean and covariance functions of the latent process. By performing functional principal component analysis to the latent process, their methods provide insights into the latent factors and correlation structures in the point process. They also proposed an empirical Bayes method to predict the latent principal component scores, which can help highlight hot areas with unusually high event rates. The latent functional data modeling in Li and Guan [57] was also previously used to model latent processes in non-Gaussian longitudinal data analysis [41,48], but the challenge in the spatio-temporal point process data analysis in Li and Guan [57] is that there are no independent replicates of the process and all data are correlated. A similar latent functional modeling approach was also adopted by [33] to jointly model grid points and response with longitudinal data.

6. Discussion

The strength of Functional Data Analysis lies in its broad applications in various domain fields. In this article, we review some recent developments in a few focused areas of FDA and their impacts on various scientific, medical, and business

applications. There are many other exciting and influential works that we have not fully expanded on, due to the limited space and our own limited expertise. For example, the optimal function-on-function regression using a Reproducing Kernel Hilbert Space approach [101], the scalar on image regression with a penalty on the total variation [110], the functional graphical models [86], and hypothesis testing in large scale functional linear regressions [116]. There have also been a lot of recent developments on functional data in a manifold [21] and Fréchet regression on random objects [85]. We believe these emerging new areas will continue to find important new applications and FDA will continue to be an active contributor to *Journal of Multivariate Analysis* in many years to come.

CRediT authorship contribution statement

Yehua Li: Main contributor for Sections 2, 5 and 6, Contributed to Section 1. **Yumou Qiu:** Main author for Section 4, Contributed to Section 1. **Yuhang Xu:** Main author for Section 3, Contributed to Section 1.

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References

[1] J. Adams, Y. Qiu, L. Posadas, K. Eskridge, G. Graef, Phenotypic trait extraction of soybean plants using deep convolutional neural networks with transfer learning, Big Data Inf. Anal. 6 (bigdia-06-003) (2021) 26–40.
[2] J. Adams, Y. Qiu, Y. Xu, J.C. Schnable, Plant segmentation by supervised machine learning methods, Plant Phenome J. 3 (1) (2020) e20001.
[3] J. Ashburner, K.J. Friston, Voxel-based morphometry—the methods, Neuroimage 11 (6) (2000) 805–821.
[4] A. Aue, D.D. Norinho, S. Hörmann, On the prediction of stationary functional time series, J. Amer. Statist. Assoc. 110 (509) (2015) 378–392.
[5] A.J. Baddeley, J. Møller, R. Waagepetersen, Non- and semi- parametric estimation of interaction in inhomogeneous point patterns, Stat. Neerl. 54 (2000) 329–350.
[6] S. Banerjee, B.P. Carlin, A.E. Gelfand, Hierarchical Modeling and Analysis for Spatial Data, Chapman and Hall/CRC, New York, 2004.
[7] R.F. Barber, M. Reimherr, T. Schill, The function-on-scalar LASSO with applications to longitudinal GWAS, Electron. J. Stat. 11 (1) (2017) 1351–1389.
[8] A. Brix, P.J. Diggle, Spatiotemporal prediction for log-Gaussian cox processes, J. R. Stat. Soc. Ser. B Stat. Methodol. 63 (2001) 823–841.
[9] A. Brix, J. Møller, Space-time multi type log Gaussian cox processes with a view to modelling weeds, Scand. J. Stat. 28 (2001) 471–488.
[10] B.A. Brumback, J.A. Rice, Smoothing spline models for the analysis of nested and crossed samples of curves, J. Amer. Statist. Assoc. 93 (1998) 961–976.
[11] P. Bühlmann, S. Van De Geer, Statistics for High-Dimensional Data: Methods, Theory and Applications, Springer, New York, 2011.
[12] T.T. Cai, P. Hall, Prediction in functional linear regression, Ann. Statist. 34 (5) (2006) 2159–2179.
[13] G. Cao, L. Wang, Simultaneous inference for the mean of repeated functional data, J. Multivariate Anal. 165 (2018) 279–295.
[14] G. Cao, L. Wang, Y. Li, L. Yang, Oracle-efficient confidence envelopes for covariance functions in dense functional data, Statist. Sinica 26 (1) (2016) 359–383.
[15] H. Cardot, P. Sarda, Estimation in generalized linear models for functional data via penalized likelihood, J. Multivariate Anal. 92 (2005) 24–41.
[16] Y. Chen, J. Goldsmith, R.T. Ogden, Variable selection in function-on-scalar regression, Stat 5 (1) (2016) 88–101.
[17] W. Chu, R. Li, M. Reimherr, Feature screening for time-varying coefficient models with ultrahigh dimensional longitudinal data, Ann. Appl. Stat. 10 (2) (2016) 596.
[18] C.M. Crainiceanu, A.-M. Staicu, C.-Z. Di, Generalized multilevel functional regression, J. Amer. Statist. Assoc. 104 (488) (2009) 1550–1561.
[19] N.A.C. Cressie, Statistics for Spatial Data, Wiley, New York, 1993.
[20] N. Cressie, C.K. Wikle, Statistics for Spatio-Temporal Data, John Wiley & Sons, Hoboken, NJ, 2015.
[21] X. Dai, H.-G. Müller, Principal component analysis for functional data on Riemannian manifolds and spheres, Ann. Statist. 46 (2018) 3334–3361.
[22] X. Dai, H.-G. Müller, W. Tao, Derivative principal component analysis for representing the time dynamics of longitudinal and functional data, Statist. Sinica 28 (3) (2018) 1583–1609.
[23] J. Dauxois, A. Pousse, Y. Romain, Asymptotic theory for the principal component analysis of a vector random function: some applications to statistical inference, J. Multivariate Anal. 12 (1982) 136–154.
[24] C. Di, C.M. Crainiceanu, B.S. Caffo, N.M. Punjabi, Multilevel functional principal component analysis, Ann. Appl. Stat. 3 (2009) 458–488.
[25] C. Di, C.M. Crainiceanu, W.S. Jank, Multilevel sparse functional principal component analysis, Stat 3 (1) (2014) 126–143.
[26] P.J. Diggle, Spatio-temporal point processes, partial likelihood, foot and mouth disease, Stat. Methods Med. Res. 15 (2006) 325–336.
[27] B. Efron, T. Hastie, I. Johnstone, R. Tibshirani, Least angle regression, Ann. Statist. 32 (2) (2004) 407–499.
[28] J. Fan, I. Gijbels, Local Polynomial Modelling and its Applications, in: Monographs on Statistics and Applied Probability, vol. 66, Routledge, 2018.
[29] Z. Fan, M. Reimherr, High-dimensional adaptive function-on-scalar regression, Econometr. Statist. 1 (2017) 167–183.
[30] J. Fan, C. Zhang, J. Zhang, Generalized likelihood ratio statistics and wilks phenomenon, Ann. Statist. (2001) 153–193.
[31] J. Friedman, T. Hastie, R. Tibshirani, Regularization paths for generalized linear models via coordinate descent, J. Stat. Softw. 33 (1) (2010) 1.
[32] A.E. Gelfand, A.M. Schmidt, S. Banerjee, C. Sirmans, Nonstationary multivariate process modeling through spatially varying coregionalization, Test 13 (2) (2004) 263–312.
[33] D. Gervini, T.J. Baur, Joint models for grid point and response processes in longitudinal and functional data, Statist. Sinica 30 (2020) 1905–1924.
[34] R. Giraldo, P. Delicado, J. Mateu, Ordinary kriging for function-valued spatial data, Environ. Ecol. Stat. 18 (3) (2011) 411–426.
[35] J. Goldsmith, X. Liu, J. Jacobson, A. Rundle, New insights into activity patterns in children, found using functional data analyses, Med. Sci. Sports Exerc. 48 (9) (2016) 1723.
[36] J. Goldsmith, V. Zipunnikov, J. Schrack, Generalized multilevel function-on-scalar regression and principal component analysis, Biometrics 71 (2015) 344–353.
[37] O. Gromenko, P. Kokoszka, L. Zhu, J. Sojka, Estimation and testing for spatially indexed curves with application to ionospheric and magnetic field trends, Ann. Appl. Stat. 6 (2) (2012) 669–696.

[38] L. Gu, L. Wang, W.K. Härdle, L. Yang, A simultaneous confidence corridor for varying coefficient regression with sparse functional data, Test 23 (4) (2014) 806–843.
[39] P. Hall, M. Hosseini-Nasab, On properties of functional principal components analysis, J. R. Stat. Soc. Ser. B Stat. Methodol. 68 (2006) 109–126.
[40] P. Hall, H.G. Müller, J.L. Wang, Properties of principal component methods for functional and longitudinal data analysis, Ann. Statist. 34 (2006) 1493–1517.
[41] P. Hall, H.G. Müller, F. Yao, Modeling sparse generalized longitudinal observations with latent Gaussian processes, J. R. Stat. Soc. Ser. B Stat. Methodol. 70 (2008) 703–723.
[42] W. Härdle, Asymptotic maximal deviation of M-smoothers, J. Multivariate Anal. 29 (1989) 163–179.
[43] T. Hastie, R. Tibshirani, Varying-coefficient models, J. R. Stat. Soc. Ser. B Stat. Methodol. 55 (4) (1993) 757–779.
[44] G. He, H.G. Müller, J.L. Wang, Functional canonical analysis for square integrable stochastic processes, J. Multivariate Anal. 85 (2003) 54–77.
[45] S. Hörmann, P. Kokoszka, Weakly dependent functional data, Ann. Statist. 38 (3) (2010) 1845–1884.
[46] L. Horváth, P. Kokoszka, Inference for Functional Data with Applications, Springer, New York, 2012.
[47] T. Hsing, R. Eubank, Theoretical Foundations of Functional Data Analysis, with an Introduction to Linear Operators, Wiley, New York, 2015.
[48] H. Huang, Y. Li, Y. Guan, Joint modeling and clustering paired generalized longitudinal trajectories with application to cocaine abuse treatment data, J. Amer. Statist. Assoc. 109 (2014) 1412–1424.
[49] J.Z. Huang, C.O. Wu, L. Zhou, Polynomial spline estimation and inference for varying coefficient models with longitudinal data, Statist. Sinica (2004) 763–788.
[50] Q. Jiang, H. Wang, Y. Xia, G. Jiang, On a principal varying coefficient model, J. Amer. Statist. Assoc. 108 (501) (2013) 228–236.
[51] N.K. Kishor, J. Morley, What factors drive the price–rent ratio for the housing market? A modified present-value analysis, J. Econom. Dynam. Control 58 (2015) 235–249.
[52] P. Kokoszka, M. Reimherr, Introduction to Functional Data Analysis, CRC Press, New York, 2017.
[53] D.R. Kowal, D.S. Matteson, D. Ruppert, A Bayesian multivariate functional dynamic linear model, J. Amer. Statist. Assoc. 112 (2017) 733–744.
[54] T. Kuenzer, S. Hörmann, P. Kokoszka, Principal component analysis of spatially indexed functions, J. Amer. Statist. Assoc. (2020) http://dx.doi.org/10.1080/01621459.2020.1732395, (in press).
[55] D. LeBauer, Data from: TERRA-REF, an open reference data set from high resolution genomics, phenomics, and imaging sensors, Dryad Dataset (2020) http://dx.doi.org/10.5061/dryad.4b8gtht99.
[56] W. Lee, M.F. Miranda, P. Rausch, V. Baladandayuthapani, M. Fazio, J.C. Downs, J.S. Morris, Bayeslan semiparametric functional mixed models for serially correlated functional data, with application to glaucoma data, J. Amer. Statist. Assoc. 114 (526) (2019) 495–513.
[57] Y. Li, Y. Guan, Functional principal component analysis of spatiotemporal point processes with applications in disease surveillance, J. Amer. Statist. Assoc. 109 (507) (2014) 1205–1215.
[58] Y. Li, T. Hsing, On rates of convergence in functional linear regression, J. Multivariate Anal. 98 (2007) 1782–1804.
[59] Y. Li, T. Hsing, Uniform convergence rates for nonparametric regression and principal component analysis in functional/longitudinal data, Ann. Statist. 38 (2010) 3321–3351.
[60] H. Li, S. Kozey Keadle, J. Staudenmayer, H. Assaad, J.Z. Huang, R.J. Carroll, Methods to assess an exercise intervention trial based on 3-level functional data, Biostatistics 16 (4) (2015) 754–771.
[61] Y. Li, D.V. Nguyen, S. Banerjee, C.M. Rhee, K. Kalantar-Zadeh, E. Kürüm, D. Şentürk, Multilevel modeling of spatially nested functional data: Spatiotemporal patterns of hospitalization rates in the US dialysis population, Stat. Med. (2021).
[62] H. Li, J. Staudenmayer, R.J. Carroll, Hierarchical functional data with mixed continuous and binary measurements, Biometrics 70 (4) (2014) 802–811.
[63] Y. Li, N. Wang, R.J. Carroll, Selecting the number of principal components in functional data, J. Amer. Statist. Assoc. 108 (2013) 1284–1294.
[64] Y. Li, N. Wang, M. Hong, N.D. Turner, J.R. Lupton, R.J. Carroll, Nonparametric estimation of correlation functions in longitudinal and spatial data, with application to colon carcinogenesis experiments, Ann. Statist. 35 (4) (2007) 1608–1643.
[65] X. Li, L. Wang, H.J. Wang, A.D.N. Initiative, Sparse learning and structure identification for ultrahigh-dimensional image-on-scalar regression, J. Amer. Statist. Assoc. (2020) 1–15.
[66] Y. Li, I. Xu, C. Liu, Functional data modeling and hypothesis testing for longitudinal alzheimer genome-wide association studies, in: Y. Zhao, D. Chen (Eds.), Modern Statistical Methods for Health Research, Springer, New York, 2021, (in press).
[67] D. Liang, H. Zhang, X. Chang, H. Huang, Modeling and regionalization of China's PM2. 5 using spatial-functional mixture models, J. Amer. Statist. Assoc. 116 (533) (2021) 116–132.
[68] Y. Liu, Y. Li, R.J. Carroll, N. Wang, Predictive functional linear models with semiparametric single-index interactions, J. Econometrics (2021) (in press).
[69] J. Liu, R. Li, R. Wu, Feature selection for varying coefficient models with ultrahigh-dimensional covariates, J. Amer. Statist. Assoc. 109 (505) (2014) 266–274.
[70] B. Liu, H.G. Müller, Estimating derivatives for samples of sparsely observed functions with application to online auction dynamics, J. Amer. Statist. Assoc. 104 (2009) 704–717.
[71] C. Liu, D.B. Rubin, The ECME algorithm: a simple extension of EM and ECM with faster monotone convergence, Biometrika 81 (4) (1994) 633–648.
[72] A. Menafoglio, O. Grujic, J. Caers, Universal kriging of functional data: Trace-variography vs cross-variography? Application to gas forecasting in unconventional shales, Spat. Statist. 15 (2016) 39–55.
[73] A. Menafoglio, G. Petris, Kriging for Hilbert-space valued random fields: The operatorial point of view, J. Multivariate Anal. 146 (2016) 84–94.
[74] C. Miao, Y. Xu, S. Liu, P.S. Schnable, J. Schnable, Increased power and accuracy of causal locus identification in time-series genome-wide association in sorghum, Plant Physiol. 183 (2020) 1898–1909.
[75] A. Mirshani, M. Reimherr, Adaptive function-on-scalar regression with a smoothing elastic net, J. Multivariate Anal. (2021) 104765.
[76] J. Møller, A.R. Syversveen, R.P. Waagepetersen, Log-gaussian cox processes, Scand. J. Stat. 25 (1998) 451–482.
[77] J. Møller, R. Waagepetersen, Modern statistics for spatial point processes, Scand. J. Stat. 34 (2007) 643–684.
[78] J.S. Morris, Functional regression, Annu. Rev. Stat. Appl. 2 (2015) 321–359.
[79] J.S. Morris, R.J. Carroll, Wavelet-based functional mixed models, J. R. Stat. Soc. Ser. B Stat. Methodol. 68 (2) (2006) 179–199.
[80] J.S. Morris, M. Vannucci, P.J. Brown, R.J. Carroll, Wavelet-based nonparametric modeling of hierarchical functions in colon carcinogenesis, J. Amer. Statist. Assoc. 98 (463) (2003) 573–583.
[81] H.-G. Müller, F. Yao, Empirical dynamics for longitudinal data, Ann. Statist. 38 (6) (2010) 3458–3486.
[82] D. Nerini, P. Monestiez, C. Manté, Cokriging for spatial functional data, J. Multivariate Anal. 101 (2) (2010) 409–418.
[83] B. Noh, A. Bandyopadhyay, W.A. Peer, E.P. Spalding, A.S. Murphy, Enhanced gravi-and phototropism in plant mdr mutants mislocalizing the auxin efflux protein PIN1, Nature 423 (6943) (2003) 999–1002.
[84] A. Parodi, M. Reimherr, Simultaneous variable selection and smoothing for high-dimensional function-on-scalar regression, Electron. J. Stat. 12 (2) (2018) 4602–4639.
[85] A. Petersen, H.-G. Müller, Fréchet regression for random objects with euclidean predictors, Ann. Statist. 47 (2019) 691–719.

[86] X. Qiao, S. Guo, G.M. James, Functional graphical models, J. Amer. Statist. Assoc. 114 (525) (2019) 211–222.
[87] J.O. Ramsay, B.W. Silverman, Functional Data Analysis, second ed., Springer-Verlag, New York, 2005.
[88] M. Reimherr, D. Nicolae, A functional data analysis approach for genetic association studies, Ann. Appl. Stat. 8 (1) (2014) 406–429.
[89] P.T. Reiss, L. Huang, M. Mennes, Fast function-on-scalar regression with penalized basis expansions, Int. J. Biostat. 6 (1) (2010).
[90] J.A. Rice, B.W. Silverman, Estimating the mean and covariance structure nonparametrically when the data are curves, J. R. Stat. Soc. Ser. B Stat. Methodol. 53 (1) (1991) 233–243.
[91] D. Ruppert, M.P. Wand, R.J. Carroll, Semiparametric Regression, Cambridge University Press, New York, 2003.
[92] F.P. Schoenberg, Multidimensional residual analysis of point process models for earthquake occurrences, J. Amer. Statist. Assoc. 98 (2003) 789–795.
[93] N. Serban, H. Jiang, Multilevel functional clustering analysis, Biometrics 68 (2012) 805–814.
[94] N. Serban, A. Staicu, R.J. Carroll, Multilevel cross-dependent binary longitudinal data, Biometrics 69 (2013) 903–913.
[95] H. Shin, S. Lee, Canonical correlation analysis for irregularly and sparsely observed functional data, J. Multivariate Anal. 134 (2015) 1–18.
[96] H. Shou, V. Zipunnikov, C.M. Crainiceanu, S. Greven, Structured functional principal component analysis, Biometrics 71 (2015) 247–257.
[97] J. Song, B. Li, Nonlinear and additive principal component analysis for functional data, J. Multivariate Anal. 181 (2021) 104675.
[98] A.-M. Staicu, C.M. Crainiceanu, R.J. Carroll, Fast methods for spatially correlated multilevel functional data, Biostatistics 11 (2) (2010) 177–194.
[99] A.-M. Staicu, C.M. Crainiceanu, D.S. Reich, D. Ruppert, Modeling functional data with spatially heterogeneous shape characteristics, Biometrics 68 (2) (2012) 331–343.
[100] M.L. Stein, Interpolation of Spatial Data: Some Theory for Kriging, Springer-Verlag, New York, 2012.
[101] X. Sun, P. Du, X. Wang, P. Ma, Optimal penalized function-on-function regression under a reproducing kernel Hilbert space framework, J. Amer. Statist. Assoc. 113 (524) (2018) 1601–1611.
[102] J. Tang, Y. Li, Y. Guan, Generalized quasi-likelihood ratio tests for semiparametric analysis of covariance models in longitudinal data, J. Amer. Statist. Assoc. 111 (2016) 736–747.
[103] Y. Tang, H.J. Wang, Z. Zhu, X. Song, A unified variable selection approach for varying coefficient models, Statist. Sinica 22 (2012) 601–628.
[104] P.M. Visscher, W.G. Hill, N.R. Wray, Heritability in the genomics era—concepts and misconceptions, Nature Rev. Genet. 9 (4) (2008) 255–266.
[105] G. Wahba, Spline Models for Observational Data, SIAM, Philadelphia, 1990.
[106] J. Wang, G. Cao, L. Wang, L. Yang, Simultaneous confidence band for stationary covariance function of dense functional data, J. Multivariate Anal. 176 (2020) 104584.
[107] J.L. Wang, J.-M. Chiou, H.G. Müller, Functional data analysis, Annu. Rev. Stat. Appl. 3 (2016) 257–295.
[108] R. Wang, Y. Qiu, Y. Zhou, Z. Liang, J.C. Schnable, A high-throughput phenotyping pipeline for image processing and functional growth curve analysis, Plant Phenomics 2020 (2020).
[109] H. Wang, P.-S. Zhong, Y. Cui, Y. Li, Unified empirical likelihood ratio tests for functional concurrent linear models and the phase transition from sparse to dense functional data, J. R. Stat. Soc. Ser. B Stat. Methodol. 80 (2) (2018) 343–364.
[110] X. Wang, H. Zhu, ADNI, Generalized scalar-on-image regression models via total variation, J. Amer. Statist. Assoc. 112 (519) (2017) 1156–1168.
[111] R. Wong, Y. Li, Z. Zhu, Partially linear functional additive models for multivariate functional data, J. Amer. Statist. Assoc. 114 (2019) 406–418.
[112] L. Xiao, L. Huang, J.A. Schrack, L. Ferrucci, V. Zipunnikov, C.M. Crainiceanu, Quantifying the lifetime circadian rhythm of physical activity: a covariate-dependent functional approach, Biostatistics 16 (2) (2015) 352–367.
[113] Y. Xu, Y. Li, D. Nettleton, Nested hierarchical functional data modeling and inference for the analysis of functional plant phenotypes, J. Amer. Statist. Assoc. 113 (522) (2018) 593–606.
[114] Y. Xu, Y. Li, Y. Qiu, Growth dynamics and heritability for plant high-throughput phenotyping studies using hierarchical functional data analysis, Biom. J. (2021).
[115] Y. Xu, Y. Qiu, J.C. Schnable, Functional modeling of plant growth dynamics, Plant Phenome J. 1 (2018) 170007.
[116] K. Xue, F. Yao, Hypothesis testing in large-scale functional linear regression, Statist. Sinica (2021) (in press).
[117] F. Yao, T. Lee, Penalized spline models for functional principal component analysis, J. R. Stat. Soc. Ser. B Stat. Methodol. 68 (2006) 3–25.
[118] F. Yao, H.G. Müller, J.L. Wang, Functional data analysis for sparse longitudinal data, J. Amer. Statist. Assoc. 100 (2005) 577–590.
[119] F. Yao, S. Sue-Chee, F. Wang, Regularized partially functional quantile regression, J. Multivariate Anal. 156 (2017) 39–56.
[120] S. Yu, G. Wang, L. Wang, C. Liu, L. Yang, Estimation and inference for generalized geoadditive models, J. Amer. Statist. Assoc. 115 (530) (2020) 761–774.
[121] C.-H. Zhang, Nearly unbiased variable selection under minimax concave penalty, Ann. Statist. 38 (2) (2010) 894—942.
[122] L. Zhang, V. Baladandayuthapani, H. Zhu, K.A. Baggerly, T. Majewski, B.A. Czerniak, J.S. Morris, Functional CAR models for large spatially correlated functional datasets, J. Amer. Statist. Assoc. 111 (514) (2016) 772–786.
[123] H. Zhang, Y. Li, Unified principal component analysis for sparse and dense functional data under spatial dependency, J. Bus. Econom. Statist. (2021) (in press).
[124] X. Zhang, J.L. Wang, From sparse to dense functional data and beyond, Ann. Statist. 44 (5) (2016) 2281–2321.
[125] L. Zhou, J.Z. Huang, R.J. Carroll, Joint modelling of paired sparse functional data using principal components, Biometrika 95 (2008) 601–619.
[126] L. Zhou, J. Huang, J.G. Martinez, A. Maity, V. Baladandayuthapani, R.J. Carroll, Reduced rank mixed effects models for spatially correlated hierarchical functional data, J. Amer. Statist. Assoc. 105 (2010) 390–400.
[127] H. Zhu, J. Fan, L. Kong, Spatially varying coefficient model for neuroimaging data with jump discontinuities, J. Amer. Statist. Assoc. 109 (507) (2014) 1084–1098.
[128] H. Zhu, R. Li, R. Zhang, H. Lian, Nonlinear functional canonical correlation analysis via distance covariance, J. Multivariate Anal. 180 (2020) 104662.
[129] V. Zipunnikov, B. Caffo, D.M. Yousem, C. Davatzikos, B.S. Schwartz, C. Crainiceanu, Multilevel functional principal component analysis for high-dimensional data, J. Comput. Graph. Statist. 20 (4) (2011) 852–873.
[130] H. Zou, The adaptive lasso and its oracle properties, J. Amer. Statist. Assoc. 101 (476) (2006) 1418–1429.